# BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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**Due Date**

Thursday, 12/04/24, 9:00pm

## Overview

### Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

### Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/Desktop/4750 Cloned Repositories/hw5-wrn25_hw5`
   Installed PlotUtils        v1.4.4
   Installed GR_jll         v0.73.18+0
   Installed HiGHS_jll        v1.12.0+0
```

```
  Installed OpenBLAS32_jll     v0.3.29+0
  Installed Measures        v0.3.3
  Installed OpenSSL        v1.6.0
  Installed MutableArithmetics   v1.6.7
  Installed Pango_jll       v1.57.0+0
  Installed FFMPEG         v0.4.5
  Installed JSON          v1.3.0
  Installed METIS_jll       v5.1.3+0
  Installed GraphRecipes      v0.5.15
  Installed DataStructures     v0.19.3
  Installed GR           v0.73.18
  Installed HiGHS         v1.20.1
  Installed StableRNGs       v1.0.4
  Installed FFMPEG_jll       v8.0.0+0
  Installed StructUtils      v2.6.0
  Installed StatsBase       v0.34.8
  Installed ForwardDiff      v1.3.0
  Installed JuMP          v1.29.3
Precompiling project...
    467.6 ms   Measures
    379.8 ms   DiffResults
    387.4 ms   StableRNGs
    589.3 ms   StructUtils
    479.9 ms   METIS_jll
    521.3 ms   OpenBLAS32_jll
    932.2 ms   ChainRulesCore
   1328.7 ms   DataStructures
    762.1 ms   ColorVectorSpace → SpecialFunctionsExt
    827.2 ms   Pango_jll
   1436.3 ms   OpenSSL
   1117.6 ms   FFMPEG_jll
    501.5 ms   StructUtils → StructUtilsTablesExt
    695.9 ms   ChainRulesCore → ChainRulesCoreSparseArraysExt
   1882.2 ms   HiGHS_jll
   3960.3 ms   MutableArithmetics
   1089.0 ms   LogExpFunctions → LogExpFunctionsChainRulesCoreExt
   2954.1 ms   ForwardDiff
   2572.4 ms   JSON
    663.7 ms   SortingAlgorithms
   1239.6 ms   SpecialFunctions → SpecialFunctionsChainRulesCoreExt
   5230.7 ms   StaticArrays
   1204.9 ms   BenchmarkTools
   2463.6 ms   FFMPEG
```

```
  1804.9 ms  StatsBase
  1231.7 ms  ArnoldiMethod
   717.2 ms  StaticArrays → StaticArraysChainRulesCoreExt
   628.7 ms  StaticArrays → StaticArraysStatisticsExt
  3092.4 ms  GR_jll
   648.2 ms  Adapt → AdaptStaticArraysExt
   931.5 ms  ForwardDiff → ForwardDiffStaticArraysExt
  7168.0 ms  PlotUtils
  1739.5 ms  GeometryTypes
  2034.9 ms  Interpolations
  2081.0 ms  PlotThemes
  3050.4 ms  Graphs
   985.4 ms  Interpolations → InterpolationsForwardDiffExt
  2635.3 ms  RecipesPipeline
  7241.8 ms  GeometryBasics
 11448.9 ms  HTTP
  1140.5 ms  NetworkLayout
  1183.6 ms  NetworkLayout → NetworkLayoutGraphsExt
  2324.1 ms  GR
  2048.6 ms  GraphRecipes
 26376.0 ms  DataFrames
 24912.1 ms  MathOptInterface
  1833.5 ms  Latexify → DataFramesExt
  1637.0 ms  MathOptIIS
  6415.4 ms  HiGHS
 11102.2 ms  JuMP
 35313.3 ms  Plots
  2928.9 ms  Plots → GeometryBasicsExt
 52 dependencies successfully precompiled in 57 seconds. 200 already precompiled.
```

```julia
using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables
```

## Problems (Total: 30 Points)

### Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF**: Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF**: Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE**: Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

| City/Facility | Landfill (km) | MRF (km) | WTE (km) |
|---|---|---|---|
| 1 | 5 | 30 | 15 |
| 2 | 15 | 25 | 10 |
| 3 | 13 | 45 | 20 |
| LF | - | 32 | 18 |
| MRF | 32 | - | 15 |
| WTE | 18 | 15 | - |

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

| Component | % of total mass | Combustion ash (%) | MRF Recycling rate (%) |
|---|---|---|---|
| Food Wastes | 15 | 8 | 0 |
| Paper & Cardboard | 40 | 7 | 55 |
| Plastics | 5 | 5 | 15 |
| Textiles | 3 | 10 | 10 |
| Rubber, Leather | 2 | 15 | 0 |
| Wood | 5 | 2 | 30 |
| Yard Wastes | 18 | 2 | 40 |
| Glass | 4 | 100 | 60 |
| Ferrous | 2 | 100 | 75 |

| Component | % of total mass | Combustion ash (%) | MRF Recycling rate (%) |
|---|---|---|---|
| Aluminum | 2 | 100 | 80 |
| Other Metal | 1 | 100 | 50 |
| Miscellaneous | 3 | 70 | 0 |

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

**Reminder**: Use `round(x; digits=n)` to report values to the appropriate precision!

### Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

### Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

### Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

### Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

### Problem 1.5

Find the optimal solution (using `JuMP` to solve the problem). Report the optimal objective value.

### Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

### Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100$MW. In period 2, the demand is projected to be $d_2 = 1200$MW, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

#### Problem 2.1

Draw a scenario tree for this problem.

#### Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

### References

List any external references consulted, including classmates.

–> Worked alone, no references besides Julia syntax pages online.